

## BAD GOISERN, Austria

WMO: 113540

Lat: 47.6425N

Lon: 13.6181E

Elev: 503

StdP: 95.43

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.0      | -9.7 | -13.8                         | 1.2 | -11.6 | -11.4 | 1.5 | -8.5 | 5.2                   | 3.3  | 3.9 | 2.9  | 0.6                   | 160  | 0.480 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.5                   | 30.9            | 20.2 | 28.9 | 19.4 | 26.9 | 18.4 | 20.8                | 28.6 | 20.0 | 27.3 | 19.2 | 25.4 | 2.6                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.4                            | 14.1 | 22.9 | 17.6 | 13.4 | 22.1 | 17.0 | 12.9 | 21.3 | 62.2          | 28.9 | 59.4 | 27.4 | 56.7 | 25.4 | 25.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 4.6           | 3.9  | 3.2 | DB                         | -14.5 | 34.4               | 3.2 | 1.4                                                | -16.8 | 35.4       | -18.7 | 36.2       | -20.4 | 37.0       | -22.7 | 38.0 | (4) |
| (5)               |      |     | WB                         | -14.4 | 22.4               | 3.3 | 1.2                                                | -16.8 | 23.3       | -18.7 | 24.0       | -20.5 | 24.6       | -22.9 | 25.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.3         | -1.3  | 0.2   | 4.5   | 9.9   | 13.8  | 17.3  | 19.2  | 18.6  | 14.3  | 9.7   | 4.3   | 0.0   |  |  |
|                                                                        | DBStd       | 8.12        | 3.75  | 4.09  | 3.94  | 3.88  | 3.84  | 3.89  | 3.50  | 3.32  | 3.52  | 3.61  | 3.94  | 3.46  |  |  |
|                                                                        | HDD10.0     | 1394        | 352   | 274   | 171   | 48    | 11    | 0     | 0     | 6     | 49    | 174   | 308   |       |  |  |
|                                                                        | HDD18.3     | 3456        | 610   | 507   | 427   | 252   | 146   | 64    | 33    | 37    | 126   | 267   | 420   | 567   |  |  |
|                                                                        | CDD10.0     | 1132        | 0     | 0     | 2     | 46    | 129   | 221   | 285   | 268   | 135   | 42    | 4     | 0     |  |  |
|                                                                        | CDD18.3     | 151         | 0     | 0     | 0     | 0     | 6     | 33    | 59    | 47    | 6     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 1476        | 0     | 0     | 0     | 18    | 109   | 340   | 549   | 412   | 47    | 1     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 453         | 0     | 0     | 0     | 2     | 24    | 105   | 185   | 131   | 7     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.0         | 0.8   | 1.0   | 1.1   | 1.2   | 1.3   | 1.2   | 1.2   | 1.1   | 0.9   | 1.0   | 0.8   | 0.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1603        | 98    | 97    | 107   | 124   | 166   | 199   | 213   | 177   | 101   | 94    | 103   | 124   |  |  |
|                                                                        | PrecMax     | 2192        | 184   | 249   | 166   | 219   | 272   | 328   | 370   | 272   | 155   | 212   | 254   | 348   |  |  |
|                                                                        | PrecMin     | 1132        | 15    | 24    | 23    | 34    | 68    | 134   | 81    | 80    | 37    | 4     | 28    | 13    |  |  |
|                                                                        | PrecStd     | 308         | 60    | 58    | 42    | 57    | 69    | 56    | 89    | 60    | 36    | 65    | 59    | 91    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.9  | 12.5  | 19.1  | 25.3  | 29.8  | 32.5  | 33.5  | 33.1  | 27.1  | 22.2  | 18.5  | 10.5  |  |  |
|                                                                        |             | MCWB        | 6.0   | 6.3   | 10.2  | 14.3  | 18.3  | 20.7  | 20.9  | 20.4  | 18.5  | 15.4  | 10.9  | 6.0   |  |  |
|                                                                        | 2%          | DB          | 5.8   | 9.5   | 16.5  | 22.5  | 26.3  | 29.8  | 30.8  | 30.3  | 24.5  | 19.2  | 15.1  | 7.2   |  |  |
|                                                                        |             | MCWB        | 4.2   | 5.2   | 9.0   | 13.2  | 16.4  | 20.0  | 20.3  | 20.0  | 17.0  | 13.4  | 9.9   | 5.5   |  |  |
|                                                                        | 5%          | DB          | 4.2   | 7.2   | 14.0  | 20.2  | 24.0  | 27.3  | 28.7  | 27.7  | 22.5  | 17.4  | 11.8  | 5.7   |  |  |
|                                                                        |             | MCWB        | 3.0   | 4.1   | 7.8   | 11.9  | 15.5  | 18.4  | 19.6  | 19.2  | 16.2  | 12.5  | 8.8   | 4.3   |  |  |
|                                                                        | 10%         | DB          | 2.9   | 5.4   | 11.5  | 18.0  | 21.7  | 25.0  | 26.6  | 25.5  | 20.6  | 15.6  | 9.6   | 4.1   |  |  |
|                                                                        |             | MCWB        | 1.9   | 3.2   | 6.6   | 10.9  | 14.2  | 17.5  | 18.7  | 18.4  | 15.5  | 12.0  | 7.8   | 3.1   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.5   | 7.2   | 10.6  | 15.0  | 19.1  | 21.6  | 22.2  | 21.6  | 19.1  | 15.8  | 11.9  | 6.9   |  |  |
|                                                                        |             | MCDB        | 9.6   | 10.5  | 17.8  | 24.1  | 26.3  | 30.1  | 30.5  | 30.1  | 25.1  | 20.5  | 16.6  | 9.2   |  |  |
|                                                                        | 2%          | WB          | 4.2   | 5.6   | 9.4   | 13.5  | 17.5  | 20.4  | 20.8  | 20.4  | 17.9  | 14.5  | 10.3  | 5.7   |  |  |
|                                                                        |             | MCDB        | 5.6   | 8.7   | 15.6  | 21.8  | 24.5  | 28.1  | 28.5  | 28.0  | 22.8  | 17.8  | 14.1  | 7.2   |  |  |
|                                                                        | 5%          | WB          | 3.1   | 4.4   | 8.2   | 12.2  | 16.2  | 19.4  | 20.1  | 19.7  | 16.9  | 13.3  | 9.1   | 4.3   |  |  |
|                                                                        |             | MCDB        | 4.0   | 6.7   | 13.1  | 19.1  | 22.5  | 25.8  | 27.4  | 26.5  | 21.2  | 16.1  | 11.5  | 5.4   |  |  |
|                                                                        | 10%         | WB          | 2.1   | 3.3   | 7.0   | 11.1  | 14.9  | 18.1  | 19.2  | 18.8  | 16.0  | 12.3  | 7.9   | 3.2   |  |  |
|                                                                        |             | MCDB        | 2.8   | 5.2   | 10.8  | 16.8  | 20.2  | 23.5  | 25.2  | 24.5  | 19.5  | 14.9  | 9.6   | 4.0   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.7   | 7.4   | 9.8   | 11.5  | 10.7  | 10.0  | 10.5  | 10.2  | 9.0   | 9.2   | 6.8   | 5.7   |  |  |
|                                                                        |             | MCDBR       | 6.6   | 10.7  | 15.5  | 16.9  | 16.5  | 15.5  | 15.1  | 14.9  | 13.0  | 12.0  | 10.8  | 7.5   |  |  |
|                                                                        | 5% WB       | MCWBR       | 4.7   | 6.6   | 8.4   | 7.8   | 7.2   | 6.0   | 5.6   | 5.7   | 6.1   | 6.5   | 6.3   | 5.4   |  |  |
|                                                                        |             | MCWBR       | 6.1   | 9.3   | 14.5  | 15.4  | 15.1  | 14.2  | 13.7  | 14.0  | 11.6  | 10.4  | 9.6   | 6.8   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.291 | 0.322 | 0.367 | 0.406 | 0.419 | 0.424 | 0.427 | 0.419 | 0.387 | 0.353 | 0.322 | 0.292 |  |  |
|                                                                        |             | taud        | 2.375 | 2.290 | 2.239 | 2.216 | 2.232 | 2.264 | 2.258 | 2.300 | 2.362 | 2.428 | 2.450 | 2.429 |  |  |
|                                                                        | Ebn at Noon |             | 789   | 831   | 844   | 844   | 846   | 842   | 833   | 824   | 817   | 791   | 746   | 745   |  |  |
|                                                                        |             | Edh at Noon | 69    | 94    | 116   | 131   | 134   | 131   | 130   | 120   | 103   | 82    | 66    | 59    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.23  | 2.17  | 3.15  | 4.12  | 4.72  | 5.20  | 5.24  | 4.66  | 3.39  | 2.35  | 1.30  | 0.99  |  |  |
|                                                                        | RadStd      |             | 0.16  | 0.27  | 0.37  | 0.51  | 0.56  | 0.53  | 0.51  | 0.55  | 0.45  | 0.24  | 0.20  | 0.15  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) Regional (95 neighbors) |  | +0.48 | N/A     | N/A    | +0.76   | +0.42 | +0.35 | N/A         | N/A     | +73     | +38     |  |  |

Nomenclature: See separate page